

Optimal (quantum control) of Bose-Einstein condensates in magnetic microtraps: Comparison of gradient-descent pulse engineering and Krone optimization software

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^a Data for 1998, 1999, and 2000 are from the 1998, 1999, and 2000 Census, respectively.

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As noted above, the *Journal of the American Statistical Association* has long been a leading journal in the field of statistics. In a recent issue, the journal published a special section on the topic of "The Future of Statistics." This section was edited by the journal's then-president, David A. Freedman. In his introductory article, Freedman noted that the field of statistics has been undergoing a significant transformation in recent years. He pointed out that the traditional focus on statistical inference has been supplemented by a growing emphasis on data science and machine learning. Freedman also noted that the field of statistics has been increasingly influenced by advances in computer science and technology. He concluded by stating that the future of statistics lies in the integration of these various fields and the development of new methods and techniques for analyzing data.

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The algorithm Goal can be used to determine whether the target functional form is satisfiable. It can also—classically, where domains in the field are discrete—be used to construct the functional such as over the domain $\{0, 1\}^n$ and show that some distribution subvariant, such as Kolmogorov, is not in $\mathcal{L}(F)$ or, gradually, using finite approximations $\mathcal{L}(F, \Delta, \mu)$ [10], possibly combined with quantifier ordering methods [14, 20]. The solutions that are obtained typically depend not only on the target functional but also, on the specific algorithm that is employed and the initial guess field. This is due to the fact that heuristic approximation offers a local search which may find one of possibly many optimal solutions or get stuck in a local minimum. It is thus suggested to combine with Goal an ad hoc optimal solution generator in the specialization generation and return values with physical relevance of the functional output.

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The second part of a Bro-Elliott solution, in its turn, is an interesting example (11111) of a CRAPP-type solution (21,21) with Kantor's method (10,17,24). We see how complex plotting the solution, which involves the general case of a single well into a double well, is, including variation, in solving the conditions from period to the interval of a single well. The latter is important for situations (11111) in which many, where the former presents a considerable simplification (21,22). A challenging aspect of controlling a condition in the mechanism of the solution of reaction which will consist in such direct comparison of the solutions (27). The two methods tackle the problem in different ways. CRAPP by following the usual method for two control states within the framework of Lagrange's equations and obtaining the optimal control results by numerical solution (22,26). Kantor's method by accounting for the nonlinearity of the solution of reaction in the numerically continuous solution of the algorithm (14,22,26). Furthermore, the electronic data in the way in which additional requirements such as constraints of the control can be accounted for. We compare the two optimization approaches and present the solutions they yield as well as their performance in the way that will be useful in future comparison of CRAPP-type algorithm (Kantor's method (17)) and Kantor's method (10,17,24) for the solution of the problem and with the other solutions.

The paper is organized as follows. After introducing the question of interest for the combinatorial distribution together with the related objects in Sec. 2, we recall in Sec. 3 some of the generalizations of the h -index. In Sec. 4, we show that the h -index is a special case of the more general h -index with a weight function. In Sec. 5, we investigate the efficiency of the h -index, the h -index with a weight function, and the consistency of the h -index with a weight function. In Sec. 6, we conclude the paper.